0 1 0 1 1 0 1 0

DESIGN & SIMULATION

8-bit Signed Binary Number Representation Using One's Complement Method

PROBLEM STATEMENT

Digital circuits store only 0s and 1s and have no native symbol for a minus sign. This project designs and simulates an 8-bit circuit that represents both positive and negative numbers using the One's Complement method, and verifies its correctness in Logisim.

TEAM

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Introduction

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1

What is signed binary representation?

A way to encode both the magnitude and the sign (positive/negative) of a number using only binary digits — no separate '+' or '-' symbol exists in hardware.

2

Why do we need One's Complement?

It lets a digital circuit represent negative numbers using the same adder/inverter hardware built for positives — subtraction becomes addition of a complemented value.

3

8-bit number representation

One bit (the MSB) is reserved as the sign bit: 0 for positive, 1 for negative. The remaining 7 bits hold the magnitude (or its complement).

One's Complement Method

THE RULE

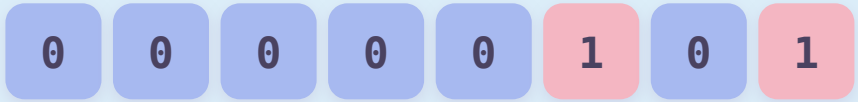
Positive numbers
Written as normal binary. No change needed.

Negative numbers
Invert every single bit (0 → 1, 1 → 0) of the positive equivalent.

KEY PROPERTY

The MSB still signals sign: 0 = positive, 1 = negative — so the sign can be read at a glance even after inversion.

+5 → POSITIVE



00000101

-5 → INVERT EVERY BIT



11111010

Design / Logic Circuit

Signal flow through the circuit, left to right

8-bit Input

Binary value entered
on 8 parallel lines

→

NOT Gates

One inverter per bit;
flips 0 ↔ 1 when active

→

Control Signal

Selects positive
(pass-through) or
negative (invert)

→

8-bit Output

Final signed value,
MSB carries the sign

HOW IT WORKS

Each of the 8 input bits feeds one NOT gate. The control input decides, per bit, whether the gate's inverted output or the original bit passes through to the output bus — inverted for negative numbers, unchanged for positive ones.

Simulation in Logisim

CIRCUIT ELEMENTS

- **Input**

  8 toggle switches feeding the bit lines

- **NOT Gates**

  8 inverters, one per bit line

- **Control Signal**

  Single switch choosing invert / pass-through

- **Output**

  8 LEDs displaying the resulting signed value

RUN 1 · CONTROL = 0 (POSITIVE)

Input +5

0 0 0 0 0 1 0 1 → Output 00000101

RUN 2 · CONTROL = 1 (NEGATIVE)

Input +5

1 1 1 1 1 0 1 0 → Output 11111010 (-5)



Conclusion

- ✓ One's Complement provides a simple way to represent signed binary numbers using only bit inversion.
- ✓ The circuit demonstrates how negative numbers can be generated from positives by inverting every bit.
- ✓ Useful for understanding digital logic fundamentals and how computers represent numbers internally.

Design and Simulation of 8-bit Signed Binary Number Representation Using One's Complement Method



Thank You

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